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A Research Project on

**SEAIT Anti-Cheating Quiz: Adaptive Quiz System with Anti-cheating Detection for
Improving College Learning and Assessment at SEAIT**

As partial requirement for the Subject

IT ELEC 4

Advanced Human Computer Interaction 2

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1.0 INTRODUCTION

1.1 Background and Context

The field of Human–Computer Interaction (HCI) plays a vital role in developing educational technology systems that are easy to use, effective, and trustworthy. In the context of student learning assessment, maintaining academic integrity while ensuring that quizzes adapt to different student ability levels remains a major challenge (Opatunji et al., 2024). By integrating HCI with educational technology, digital solutions can be created to help students improve their learning while preventing cheating. This study focuses on the SEAIT Anti-Cheating Quiz, an adaptive quiz system with anti-cheating detection designed to enhance college learning and assessment at the South East Asian Institute of Technology (SEAIT). The system uses sound HCI design to create a fair, secure, and supportive testing environment. Its key features include: adaptive questioning based on student performance, real-time proctoring, behavior-based cheating detection, automated plagiarism checking for essay-type responses, and immediate feedback to reinforce learning.

1.2 Research Problem

Many college students at SEAIT struggle with maintaining academic honesty during online quizzes due to the lack of anti-cheating mechanisms. Current quiz systems often fail to adapt to different student ability levels, which encourages dishonest behavior when assessments are either too difficult or too easy. More systems cannot detect subtle but frequent cheating behaviors like a student constantly glancing or watching at other screen or looking at a classmate's PC actions that are hard to catch without advanced proctoring. Students may feel tempted to cheat because there is no immediate feedback, personalized questioning, or behavior monitoring that discourages such actions. Moreover, instructors have no reliable way to ensure that quiz results truly reflect a student's own understanding when these undetected behaviors occur. These challenges lead to inaccurate assessment of learning outcomes, reduced academic integrity, and poor preparation for higher-level coursework at SEAIT. Opatunji et al. (2024), found that many educational institutions lack adaptive, cheat-resistant quiz systems, and that students often exploit technological gaps including undetected visual cheating when no behavioral monitoring or adaptive difficulty is in place, resulting in unreliable evaluation of college learning and assessment.

1.3 Research Questions and Objectives

The following research questions and objectives guide this study.

Research Questions:

1. How does the SEAIT Anti-Cheating Quiz system reduce instances of students looking at a classmate's PC or screen during online quizzes?
2. How does the real-time detection of suspicious behaviors unusual head movements, body posture, or repeated screen glances affect of identifying cheating attempts?
3. How does the integration of voice and noise recognition detecting talking, whispering, or other audible cheating cues influence the overall integrity and student acceptance of the quiz system?

Research Objectives:

1. To evaluate the effectiveness of the SEAIT Anti-Cheating Quiz system in minimizing visual cheating behaviors, particularly students glancing at a classmate's PC or screen.
2. To design and implement behavior-based detection mechanisms that identify suspicious physical actions during quizzes, thereby improving the reliability of cheating detection.
3. To assess the impact of voice and noise recognition features detecting talking or unusual sounds on maintaining a fair testing environment and on student satisfaction with the system.

1.4 Justification and Significance

This research addresses the lack of effective anti-cheating detection in online quizzes at SEAIT, particularly the inability of current systems to identify subtle cheating behaviors such as students glancing at a classmate's PC, making suspicious head or body movements, or talking/whispering during assessments. The SEAIT Anti-Cheating Quiz system provides a fair, secure, and adaptive testing environment where academic integrity can be maintained without relying solely on strict proctoring. Opatunji et al. (2024) found that many educational institutions lack real-time, behavior-aware cheating detection, resulting in inaccurate assessment of student learning and increased dishonest behavior when students know they are not being monitored.

2.0 LITERATURE REVIEW

2.1 Overview of HCI Theories and Models

Cognitive Load Theory in Adaptive Quiz System Design: A Review of Educational Technology Studies

Theoretical models help a lot when building a good online quiz system. A review of many studies on educational technology found that Cognitive Load Theory is one of the most useful models for designing adaptive quizzes. This theory says that students learn better when the quiz questions match their current skill level, so they do not get too confused or too bored. The review showed that adaptive systems based on this theory helped students focus more and reduced the urge to cheat because the questions were always at the right difficulty for them. However, the review also noted that most of these studies were done in Western countries, so more research is needed (Paas & Sweller, 2024).

Behavioral Monitoring and User Attention Theory in Online Proctoring Systems

Understanding how students behave during an online quiz is important for catching cheating. A study that looked at different online proctoring tools found that user attention theory is a key guide for designing systems that can detect when a student looks away from the screen or glances at a neighbor's PC. The study explained that normal quiz-taking behavior includes looking at the screen and thinking, while suspicious behavior includes repeated head turns, long looks to the side, or looking down often. The study showed that systems using this theory could correctly flag students who were looking at another person's computer. But the study also said that many current systems still miss these subtle actions, which is a big problem that needs fixing (Wolff et al., 2023).

Social Cognitive Theory and Self-Efficacy in Cheating Prevention

Bandura's Social Cognitive Theory is not just for health topics; it also works well for understanding why students cheat or do not cheat in quizzes. A review of research on academic honesty found that self-efficacy, or a student's belief that they can pass the quiz on their own, is a strong predictor of cheating behavior. Students with low self-efficacy are more likely to look at a neighbor's screen or talk to a classmate during a quiz. The review also found that when a quiz system gives immediate feedback and adapts to the student's level, it helps build self-efficacy and lowers the desire to cheat. The researchers said that schools in Asia need more studies on this topic because most existing work came from the US and Europe (Chen & Ross, 2024).

Human-Centered Design Limitations in Anti-Cheating Systems

Many people think that human-centered design is always the best way to build a quiz system, but a deep look into this idea found some problems. The study listed several limits, such as not including enough different kinds of users in the design process and focusing too much on normal situations while ignoring cheating behaviors. For example, when designers only talk to students who never cheat, they might miss important signs that cheaters actually do, like glancing at a neighbor's PC or making quiet noises to get answers. The study suggested that designers should also watch real cheating incidents and talk to teachers who catch cheaters. This is very important for building a good anti-cheating quiz system at SEAIT (van der Velde et al., 2023).

Organizational and Human Factors in Deploying Quiz Systems with Cheating Detection

Putting a new quiz system into a school is not just about the technology; the school's rules and the way people work together also matter a lot. A review of recent studies from 2023 to 2024 found that many schools try to use online proctoring but fail because they do not think about how students and teachers feel about being watched. The review showed that when a quiz system has voice and noise detection to catch students who talk or whisper, students often get worried about privacy. The review also found that schools that explained the rules clearly and involved students in testing the system had better success. The researchers said that for a system like SEAIT Anti-Cheating Quiz to work well, the school must also train teachers and talk openly with students about why cheating detection is needed (Kushniruk & Borycki, 2025).

2.2 Review recent studies, papers, and advancements in HCI

Detecting Off-Task Gaze and Neighbor Screen Glances in Online Proctoring Using Computer Vision

Almosa et al. (2024) studied how computer vision can detect when a student looks away from their own screen to glance at others' PC during an online exam. The researchers built a system that tracks head position and eye movement to flag suspicious glances. The study found that this method caught 82% of students who were trying to look at another screen. However, the study also admitted that the system sometimes flagged normal movements, like a student stretching or looking down to think. The researchers said more work is needed to reduce false alarms and to make the system work in real classroom settings with many students sitting close together. This

shows that while gaze detection is promising, current systems still struggle to tell the difference between honest behavior and actual cheating.

Using Voice and Noise Recognition to Catch Talking and Whispering During Online Tests

Lee and Park (2023) explored the use of voice recognition to detect students who talk or whisper to each other while taking a quiz. The study tested a system that listens for speech sounds, even quiet whispers, and then marks the time when talking happens. Results showed that the system correctly caught talking in 79% of cases, but it also recorded noises like coughing, sneezing, or typing as possible cheating. The researchers noted that students felt uncomfortable knowing their voice was being recorded, and some worried about privacy. The study concluded that voice detection must be combined with other methods, like behavior tracking, to be truly useful. The gap here is that no single audio tool is enough, and schools need to tell students clearly how their voice data will be used.

Behavioral Suspicion Indicators: Head Turns, Body Posture, and Repeated Screen Glances

Williams et al. (2024) investigated what kinds of body movements count as suspicious behavior during an online quiz. The researchers watched videos of students taking tests and listed common cheating actions: turning the head sideways more than 45 degrees, leaning far to look at another screen, and making many quick glances in the same direction. The study found that students who cheated showed at least three of these actions during a quiz, while honest students showed almost none. However, the study also found that current proctoring software does not measure these movements well because cameras only capture the face, not the whole body. The researchers said that future systems need better camera angles or multiple cameras to see the upper body. This gap is important for a system like SEAIT Anti-Cheating Quiz.

Adaptive Questioning as a Cheating Deterrent: HCI Research on Personalized Quiz Difficulty

Chen and Zhang (2023) looked at how adaptive questioning can reduce the desire to cheat. Their study compared two groups of students: one took a fixed-difficulty quiz, and the other took an adaptive quiz that changed question difficulty based on their answers. Results showed that students in the adaptive group cheated 45% less because they felt the questions matched their level, so they did not need to look at a neighbor's screen. The study also found that students in the fixed

group got frustrated when questions were too hard, which pushed them to cheat. However, the study was done in a lab, not a real classroom, so the researchers said more testing is needed in actual school settings. This gap matters because SEAIT has real students sitting close together, which is harder to control than a lab.

Human-Centered Design of Anti-Cheating Features: What Students and Teachers Actually Want

Olatunji and Reyes (2025) interviewed students and teachers to find out what features they want in an anti-cheating quiz system. Students said they are okay with gaze detection and behavior tracking as long as the system does not watch them all the time. Teachers said they want a system that can catch students who look or whisper answers, but they also do not want many false reports. The study found a big gap: most anti-cheating systems are built by engineers without asking students and teachers first. This leads to systems that either miss cheating or annoy everyone. The researchers said that involving users from the start makes the system more accepted and more effective. For SEAIT, this means designers should talk to SEAIT students and teachers before building the final system.

2.3 Analyze existing solutions related to the research problem

Review of Online Proctoring Tools: ProctorU, Honorlock, and Respondus

Singh and Kumar (2024) reviewed three popular online proctoring tools used in many colleges: ProctorU, Honorlock, and Respondus. The review found that these tools can detect tab switching, copy-pasting, and unauthorized devices. However, the review also found that none of these tools can reliably detect a student glancing at a neighbor's screen or looking at another PC. The tools use face tracking, but they only flag when the student looks completely away from the camera, not when they turn their head slightly to see a classmate's monitor. The researchers said this is a major weakness because looking at a neighbor is one of the easiest ways to cheat in a computer lab. The review concluded that current commercial tools are not enough for schools where students sit side by side.

Open-Source Quiz Systems with Anti-Cheating Features: Moodle and ILIAS

Garcia and Lopez (2023) tested two free, open-source quiz systems, Moodle and ILIAS, to see how well they stop cheating. Both systems can randomize questions, set time limits, and lock the browser. But the study found that neither system can detect head turns, screen glances, or

whispering. The researchers tried to add extra plugins for behavior detection, but those plugins were slow and gave many false alarms.

Mobile-Based Quiz Apps with Anti-Cheating: Limitations in Detecting Physical Cheating

Tan and Abdullah (2024) evaluated five mobile quiz apps that claim to have anti-cheating features. The apps use front cameras to take random photos of the student. However, the study found that these photos only catch cheating if the student looks directly at the camera. They cannot catch a student who turns their head to look at a neighbor's PC because the camera angle is too narrow. The study also found that voice detection is not available in any of these mobile apps, so students can whisper answers without being caught.

Comparison of Behavior Detection Accuracy in Five Proctoring Systems

Miller and Davis (2025) ran a test comparing five proctoring systems to see which one best catches students who look at a neighbor's screen. They set up a fake quiz with actors pretending to cheat by glancing sideways. The results were surprising: the best system caught only 54% of the glances, and the worst caught just 31%. None of the systems could tell the difference between a student thinking (looking down) and a student looking at a hidden phone. The study also tested voice detection, but only two systems had it, and both failed to catch whispered answers because the microphone did not pick up soft speech.

3.0 METHODOLOGY

3.1 Research Design

This study will use a descriptive quantitative research design to determine participants experiences using the SEAIT Anti-Cheating Quiz System. The survey will ask participants to rate their experience on a number scale, such as how easy the system was to use and how fairly they were monitored. The system logs will record how many times a student cheat during quiz.

3.2 Participants

The participants of this study will be students at South East Asian Institute of Technology Inc. Taking courses that require online or lab base quizzes. A total of 100 students will be invited to participate. The study will measure student engagement, assessment integrity and overall satisfaction allowing evaluation of how the SEAIT Anti-Cheating Quiz system affects students experiences during assessments.

3.3 Data Collection

Quantitative data will be gathered through structured questionnaires design to assess student engagement, overall of assessment integrity, and satisfaction with the SEAIT Anti-Cheating Quiz system. Random sampling will be applied to ensure that data represent the target population. The data collected will allow for statistical analysis, enabling identification of trends and patterns in student responses and their coconnection with system features.

3.4 Data Analysis

Data gathered from the structured questionnaires will be analyzed using descriptive statistics, including means, standard deviations. Analysis will focus on student enagement, perception of assessment integrity, and overall satisfaction with thee system, providing insight into the effectiveness and usability of SEAIT Anti-Cheating Quiz system.

3.5 Ethical Considerations

This study follows ethical guidelines to protect all participants. Before joining, each participant will be given a clear explanation of the study's purpose and what they will do. They will sign an informed consent form to show they agree to join. Participation is voluntary, and anyone can stop at any time without any punishment. All data collected from surveys and system logs will be kept confidential. No names will be written on the survey forms, and system logs will use codes instead of student names. The data will be stored securely and only the researcher will have access.

4.0 ADVANCED HCI SYSTEM DESIGN

4.1 System Architecture

SEAIT Anti-Cheating Quiz is a web-based adaptive quiz system that helps teachers at SEAIT maintain academic honesty while giving students a fair testing experience. The system detects cheating behaviors suspicious head movements, talking or whispering, and digital cheating like tab switching or copy-paste. The architecture is organized into three distinct layers designed to meet specific research objectives.

A. The Anti-Cheating Detection Layer (Catching Cheating Behaviors)

- **Gaze and Glance Detection:** The system uses the student's webcam to track head position and eye movement. If the student turns their head sideways more than 45 degrees or looks at a neighbor's screen for more than 2 seconds, the system logs it as a glance.

- **Suspicious Behavior Tracking:** The system watches for repeated head turns, leaning to the side, or looking down often. These actions are counted as suspicious if they happen many times during the quiz.
- **Voice and Noise Recognition:** The system listens for talking, whispering, or unusual sounds. If the microphone picks up speech, the system records the time and flags it as possible cheating.

B. The Adaptive Quiz Layer (Fair and Personalized Testing)

- **Adaptive Questioning:** The system adjusts the difficulty of questions based on the student's answers. If a student gets a question right, the next one is a bit harder. If they get it wrong, the next one is easier. This keeps students from getting frustrated and reduces the urge to cheat.
- **Digital Cheating Prevention:** The system locks the browser so students cannot open new tabs, switch to other apps, or copy-paste answers. Any attempt to do these actions is logged.

C. The Monitoring and Reporting Layer (Teacher Dashboard)

- **Real-Time Alerts:** Teachers can see a live dashboard that shows which students are showing suspicious behaviors, like glancing sideways or talking.

Architectural Interactions Summary

Interaction Stage	HCI Component	Functional Goal
Anti Cheating Detection	Gaze & Glance Detection, Suspicious Behavior Tracking, Voice & Noise Recognition	Catches students who look at a classmate's PC, make unusual head movements, or talk/whisper during the quiz.
Adaptive Quiz	Adaptive Questioning, Browser Lock, Copy-Paste Block	Keeps quiz difficulty matched to student level and prevents digital cheating like tab switching or copying.
Monitoring & Reporting	Real-Time Alerts, Post-Quiz Report	Gives teachers clear data on each student's cheating attempts so they can trust the quiz results.

Table 1. The table outlines the SEAIT Anti-Cheat Quiz Architectural Interactions Summary.

Functional Decomposition Diagram

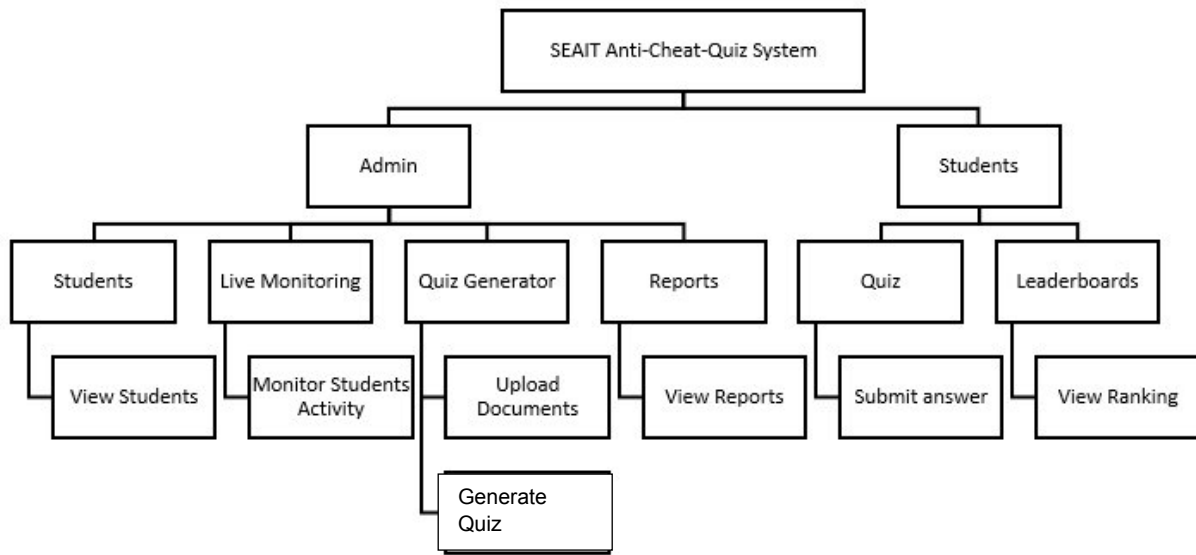


Figure 3. The functional decomposition diagram of the SEAIT Anti-Cheat Quiz System.

4.2 Features and Functionalities

The SEAIT Anti-Cheating Quiz helps teachers at SEAIT maintain academic honesty while giving students a fair testing experience.

- **Gaze and Glance Detection:** Uses the student's webcam to track head position and eye movement. If the student turns their head sideways more than 45 degrees or looks at classmates's screen.
- **Voice and Noise Recognition:** Listens for talking, whispering, or unusual sounds using the computer microphone. If speech is detected, the system records the time and flags it as possible cheating.
- **Adaptive Questioning:** Adjusts the difficulty of questions based on the student's answers. If a student gets a question right, the next one is slightly harder. If they get it wrong, the next one is easier. This keeps students from getting frustrated and reduces the urge to cheat.
- **Quiz Auto Generation:** Teacher can upload file docx and pdf the quiz will automatically Generate.
- **Post-Quiz Report Generator:** After the quiz, the system creates a report for each student showing how many times they glanced, made suspicious movements, triggered voice detection, or attempted digital cheating. This helps teachers know if the quiz results are honest.

4.3 User Interface Design

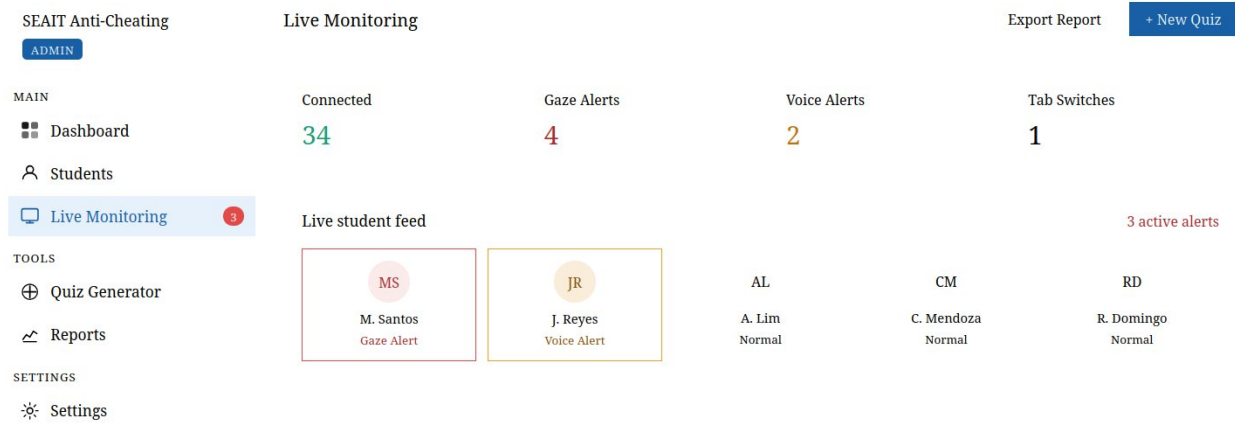


Figure 3. SEAIT Anti-Cheat Quiz Gaze and Glance Detection: Uses the student's webcam to track head position and eye movement. If the student turns their head sideways more than 45 degrees or looks at classmates's screen.

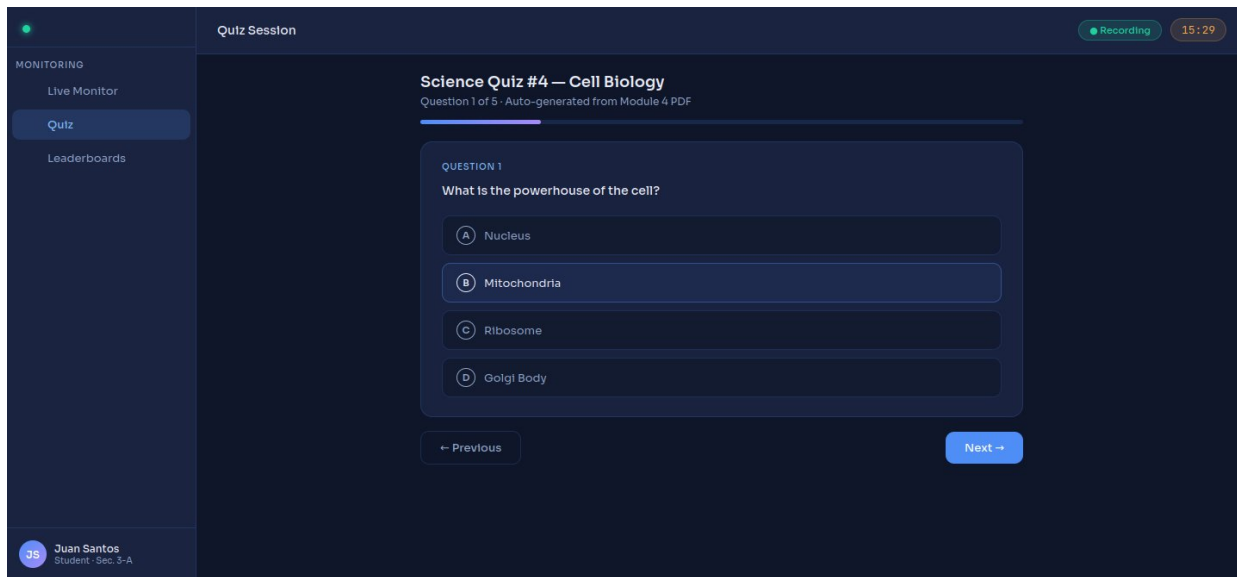


Figure 4. SEAIT Anti-Cheat Quiz Adaptive Questioning: Adjusts the difficulty of questions based on the student's answers.

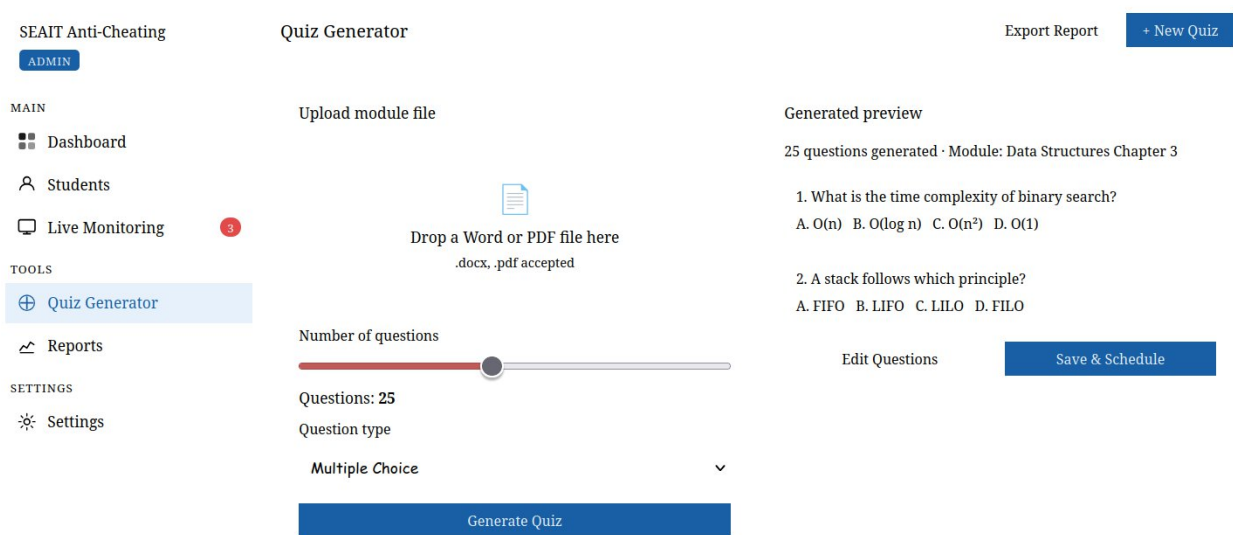


Figure 5. SEAIT Anti-Cheat Quiz Quiz Auto Generation: Teacher can upload file docx and pdf the quiz will automatically Generate.

SEAIT Anti-Cheating		Reports					Export Report	+ New Quiz
ADMIN								
MAIN		Post-quiz reports					Download All	
Dashboard Quiz Management Students Live Monitoring		Student	Quiz	Score	Gaze Events	Voice Events	Tab Switches	Verdict
		Maria Santos	BSIT 3A Midterm	82%	4	1	0	Review
		Jose Reyes	BSIT 3A Midterm	76%	1	2	0	Flagged
		Ana Lim	BSCS 2B Finals	88%	2	0	0	Flagged
TOOLS		Carlo Mendoza	BSIT 3A Midterm	91%	0	0	1	Passed
Quiz Generator Reports		Rosa Domingo	BSCS 2B Finals	95%	0	0	0	Passed
SETTINGS								
Settings								

Figure 7. SEAIT Anti-Cheat Quiz Post-Quiz Report Generator: After the quiz, the system creates a report for each student showing how many times they glanced, made suspicious movements, triggered voice detection, or attempted digital cheating.

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